

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K₄ Fe (CN)₆. After filtering, the precipitate is dried and weighed. The reactions which occur are $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$ If a 2 gram sample of rock yields 0.969 gram of Zn₂Fe(CN)₆, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn₂ Fe (CN)₆ = 342.6.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<think>

- We know that Zn in Zn₂Fe(CN)₆ comes from ZnS.
- The ratio of Zn to Zn₂Fe(CN)₆ = 1 : 1 (from the balanced equation).
- So , 1 mol of Zn gives 1 mol of Zn₂Fe(CN)₆.
- Mass of Zn = (mass of Zn₂Fe(CN)₆) * (atomic weight of Zn / molecular weight of Zn₂Fe(CN)₆).
- Mass of Zn = 0 . 9 6 9 g * (6 5 . 4 g/mol / 3 4 2 . 6 g/mol) = 0 . 1 7 4 g .
- % of Zn in the sample = (mass of Zn / mass of sample) * 1 0 0 .
- % of Zn = (0 . 1 7 4 g / 2 g) * 1 0 0 = 8 . 7 % âˆ’ 1 9 % .

</think>

<answer>

D . 1 9 % Zn

</answer>